Network Analysis for Historical Music Research: Deep-Mapping Musical Space in Milan (1958–1962)

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Abstract

In this paper, I present the results of my PhD research, which investigates the music scene of Milan during the "economic miracle" (1958–1962). The project explores how the configuration of urban space may have influenced Milanese music production, which was characterized by an unusual interconnection between genres. My work adopts a hybrid *deep-mapping* approach, integrating computational methods (geospatial and network analysis) with field-based qualitative research. Through the manual collection of listings from the newspaper *Il Giorno*, I created a dataset of 8,288 music performances. Despite its richness and scale, however, this historical dataset presents a clear asymmetry, as two venue types out of eight accounted for 93% of recorded performances.
By applying a force-based spatialization algorithm to the relationships between composers and performers (Figure 1), I was able to identify the overall structure of genre connections within the live music scene. The topology of the graph shows how classical and popular music were bridged through scenic music practices. Consistently, theatres display the highest values of betweenness centrality, which in this context indicates their role in connecting otherwise distant genres.

At the same time, field research involving musicians active in the scene added important nuance to the dataset. Interviews, in which interactive maps were used as memory triggers, revealed that the performances reported in newspaper listings represented only a portion of the overall live music activity. While theatre performances appear with continuity, nightclubs remain minimally represented despite hosting a much higher volume of events. With this new awareness, I returned to quantitative analysis. I transformed the "performers–spaces" graph into a monopartite "spaces–spaces" network, reducing the impact of data asymmetry. I then geolocated this network within an interactive map (Figure 2), showing that venues with the highest betweenness centrality values did not belong to the same venue type, but were all concentrated in the central area of the city. This suggests that the centrality of Milanese space created the conditions for cross-contamination between distant genres.

This research highlights the importance of hybrid approaches in historical network analysis. The combination of multiple points of observation created opportunities for cross-validation and, most importantly, for a productive interpretation of historical data biases. In this sense, the co-creation of knowledge through a deep-mapping approach proved crucial in shaping quantitative analysis strategies.

Finally, I address future perspectives by presenting an interactive web application prototype

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designed to visualize temporally dynamic force-based networks (Figure 3). This additional point of observation sheds light on the stability of urban performance networks, revealing the homeostatic condition of a live music ecosystem. Moreover, as the research suggests, such applications can support not only analytical processes but also future field-based enquiries.

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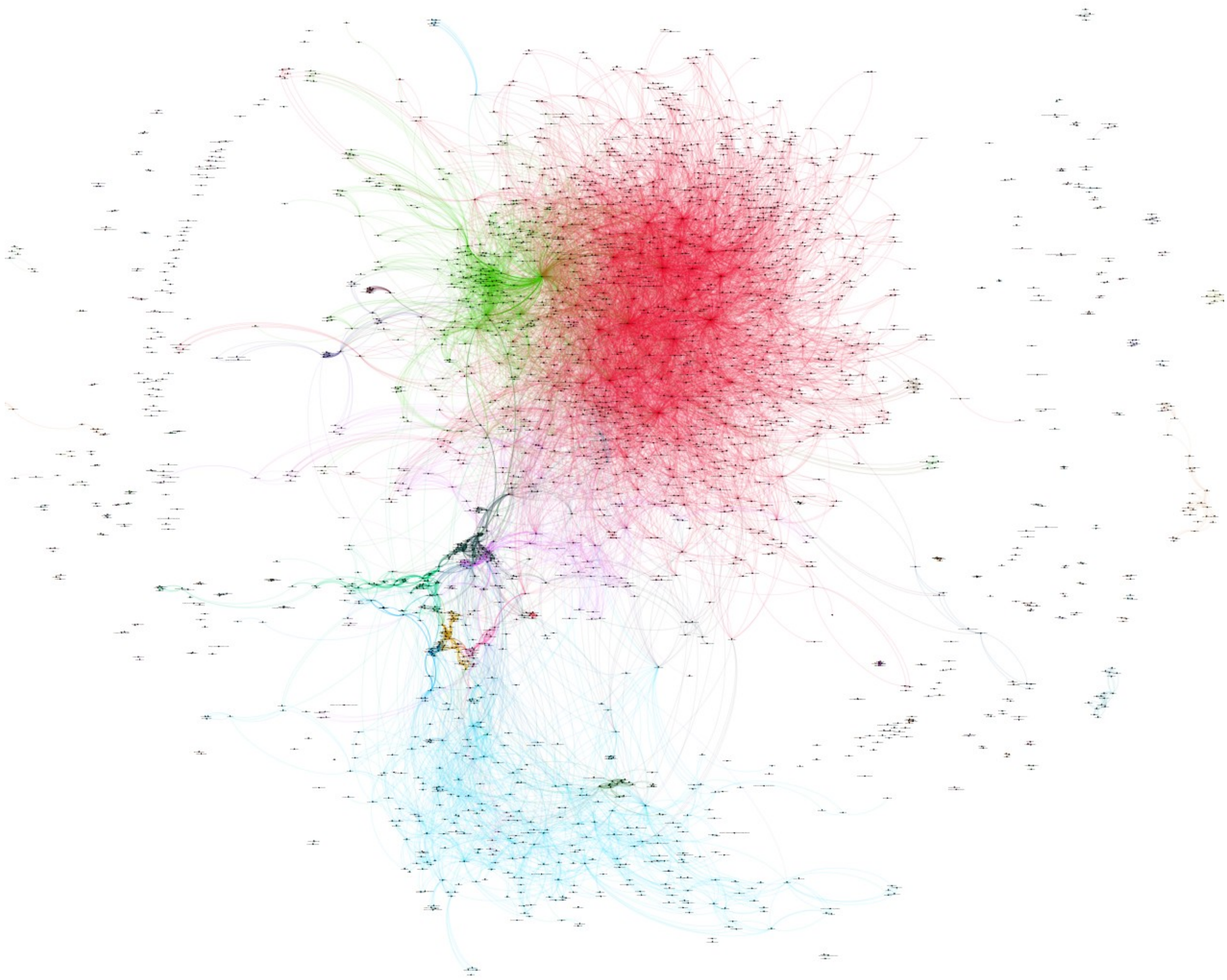

Figure 1. Composers-performers graph.

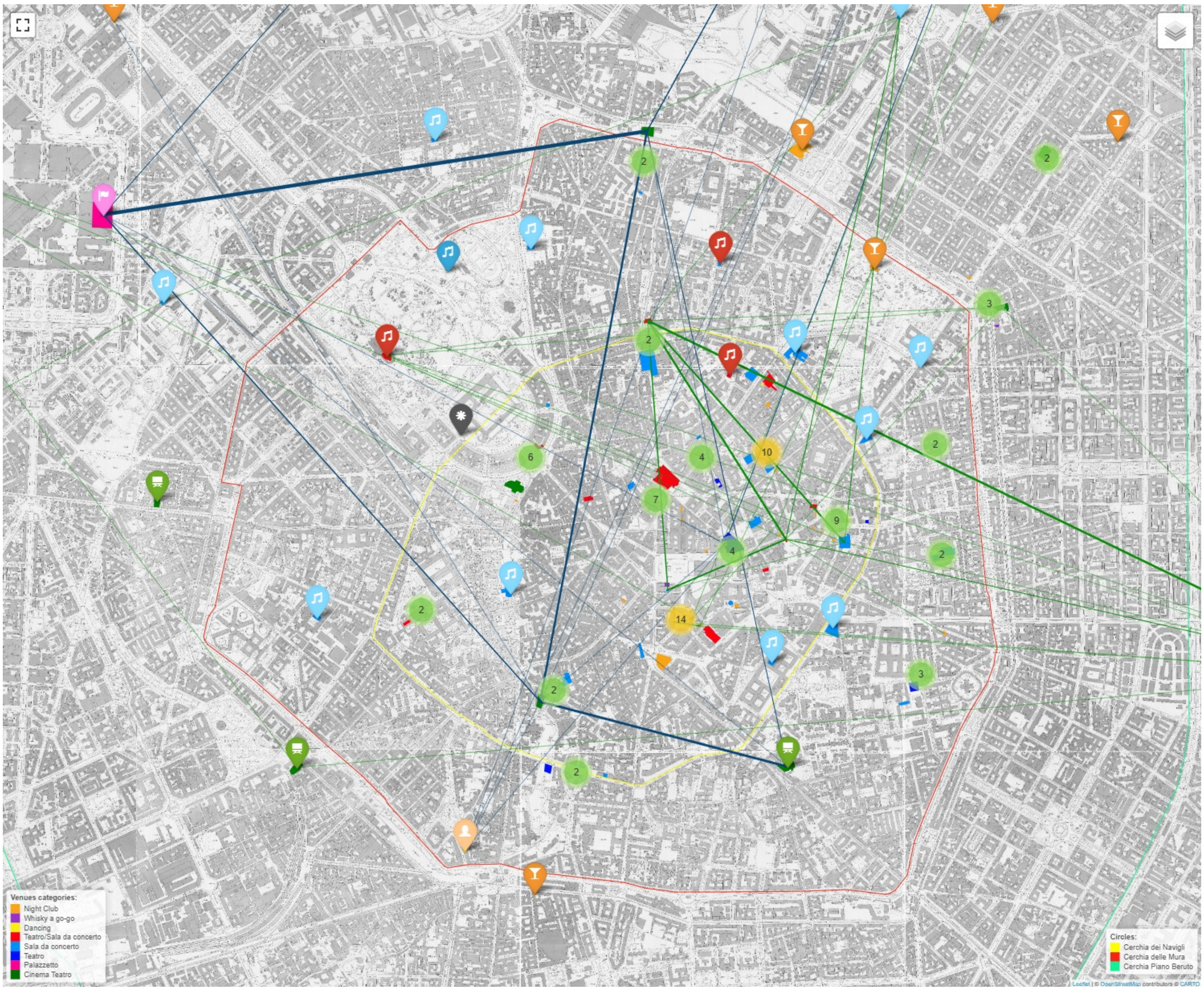

Figure 2. Performance circuits map.

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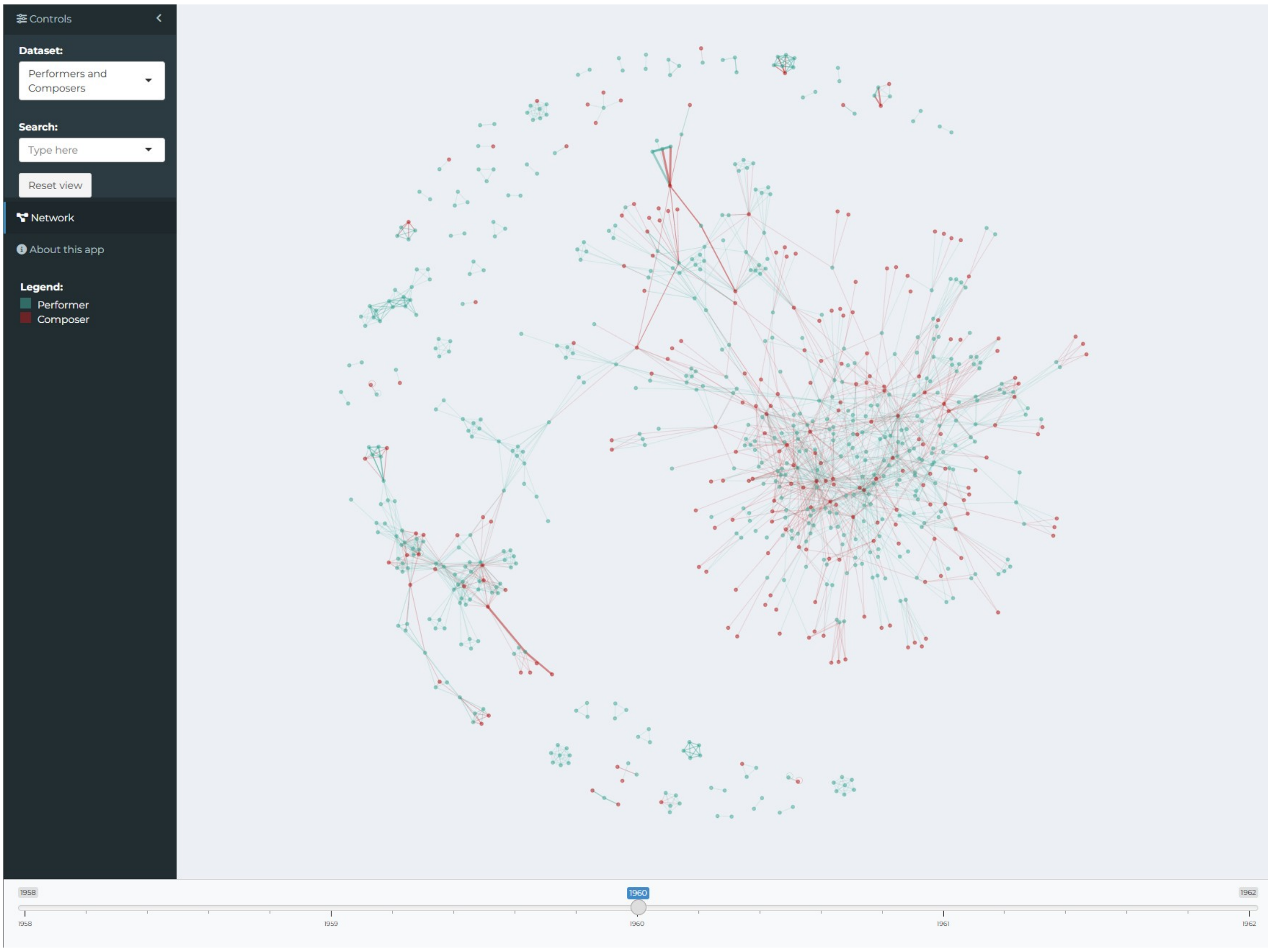

Figure 3. Time-based application.